

10 Levels of Writing Python Functions

Level 1: Basic Function

```
def my_function():  
    print("Hello, world!")  
#clcoding.com
```

Level 2: Function with Parameters

```
def greet(name):  
    print(f"Hello, {name}!")  
  
greet("Alice")  
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```

Hello, Alice!

Level 3: Return Values

```
def add(a, b):  
    return a + b  
  
result = add(3, 5)  
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```

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Level 3: Return Values

```
def add(a, b):  
    return a + b  
  
result = add(3, 5)  
#clcoding.com
```

Level 4: Default Parameters

```
def greet(name="Guest"):  
    print(f"Hello, {name}!")  
  
greet() # Will print "Hello, Guest!"  
#clcoding.com
```

Hello, Guest!

Level 5: Docstrings

```
def add(a, b):  
    """  
    This function adds two numbers.  
    """  
    return a + b  
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```

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Level 6: Variable Scope

```
global_var = 10

def some_function():
    local_var = 5
    print(global_var + local_var)

some_function()
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```

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Level 7: Recursion

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)
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```

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Level 8: Lambda Functions

```
double = lambda x: x * 2
result = double(3)
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```

Level 9: Function Decorators

```
def my_decorator(func):
    def wrapper():
        print("Something is happening before the function is called.")
        func()
        print("Something is happening after the function is called.")
    return wrapper

@my_decorator
def say_hello():
    print("Hello!")

say_hello()
#clcoding.com
```

Something is happening before the function is called.
Hello!
Something is happening after the function is called.

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Level 9: Function Decorators

```
def my_decorator(func):  
    def wrapper():  
        print("Something is happening before the function is called.")  
        func()  
        print("Something is happening after the function is called.")  
    return wrapper  
  
@my_decorator  
def say_hello():  
    print("Hello!")  
  
say_hello()  
#clcoding.com
```

Something is happening before the function is called.
Hello!
Something is happening after the function is called.

Level 10: Advanced Functions

```
def apply(func, x):  
    return func(x)  
  
def square(x):  
    return x ** 2  
  
result = apply(square, 5)  
#clcoding.com
```